SMART CITY TRAFFIC FORECASTING PROJECT REPORT

Title:

Forecasting of Smart City Traffic Patterns Using Machine Learning

1. Introduction

With the rapid growth of urban populations, traffic congestion has become a major issue in modern cities. The concept of a smart city focuses on using technology and data to improve urban services and infrastructure. One of the key challenges in smart cities is efficient traffic management.

This project aims to analyze and forecast traffic patterns at different junctions using data science and machine learning techniques.

Accurate traffic prediction can help authorities plan better road infrastructure, optimize signal timings, and reduce congestion.

2. Objectives

To analyze traffic patterns at different junctions

To identify peak traffic hours and congestion trends

To build a predictive model for traffic forecasting

To assist in smart city traffic management planning

3. Dataset Description

The dataset used in this project is taken from Kaggle and contains traffic data collected from four junctions in a city.

Dataset Features:

DateTime – Timestamp of traffic data

Junction – Junction number (1 to 4)

Vehicles – Number of vehicles passing

4. Methodology

4.1 Data Preprocessing

Converted DateTime into proper format

Sorted data chronologically

Checked and handled missing values

4.2 Feature Engineering

Extracted important features such as:

Hour

Day of the week

Month

Weekend indicator

Peak hour indicator

These features help in improving prediction accuracy.

4.3 Data Visualization

Plotted traffic trends over time

Compared traffic across junctions

Analyzed hourly traffic distribution

5. Model Development

Two machine learning models were used:

5.1 Linear Regression

Simple baseline model

Used to understand basic relationships

5.2 Random Forest Regression

Advanced model

Handles non-linear patterns better

Provided better prediction accuracy

6. Model Evaluation

The models were evaluated using:

Mean Absolute Error (MAE)

Root Mean Square Error (RMSE)

Result:

Random Forest performed better than Linear Regression

Provided more accurate traffic predictions

7. Results and Analysis

Traffic is highest during morning (7–10 AM) and evening (5–8 PM)

Weekdays have more traffic than weekends

Some junctions experience consistently higher traffic

Peak hour and junction are the most important features

8. Applications

Smart traffic signal control

Road infrastructure planning

Traffic congestion reduction

Real-time traffic monitoring systems

9. Challenges Faced

Handling time-based data

Selecting the right features

Improving model accuracy

Understanding real-world traffic patterns

10. Conclusion

This project successfully analyzed and predicted traffic patterns using machine learning techniques. The use of Random Forest improved prediction accuracy and provided useful insights into traffic behavior.

The findings can help in designing efficient smart city traffic systems.

11. Future Scope

Use advanced models like ARIMA or LSTM

Integrate real-time traffic data

Build a live traffic dashboard

Include weather and event data for better predictions

12. Tools&Technologies Used

jupyter Lab/anaconda

Python

Pandas

NumPy

Matplotlib

Scikit-learn

13. References

Kaggle Traffic Dataset

Upskill campus videos and documentation

Python Documentation

Machine Learning Tutorials